Lab 5: Phasors - Part 1

ECE2200L Section 04, Fall 2026

Matthew Ponciano, Eli Pantoja

September 28, 2026 - October 5, 2026

No 1.

Step 1. Use Phasor Transformation notation to add the two sinusoidal functions shown below:

$$i(t) = \left[\underbrace{5\cos(300t + 36.87°)}_{A} + \underbrace{10\cos(300t - 53.13°)}_{B} \right] \text{ A}$$

$$\mathbf{A} = 5\angle 36.87°$$
$$\mathbf{B} = 10\angle -53.13°$$

$$\begin{aligned}\mathbf{A} &= 5[\cos(36.87°) + j\sin(36.87°)] \\ &\approx 4 + j3 \text{ A}\end{aligned}$$

$$\begin{aligned}\mathbf{B} &= 10[\cos(-53.13°) + j\sin(-53.13°)] \\ &\approx 6 - j8 \text{ A}\end{aligned}$$

$$\mathbf{I} = \mathbf{A} + \mathbf{B} = (4 + 6) + j(3 - 8)$$
$$\mathbf{I} = 10 - j5$$

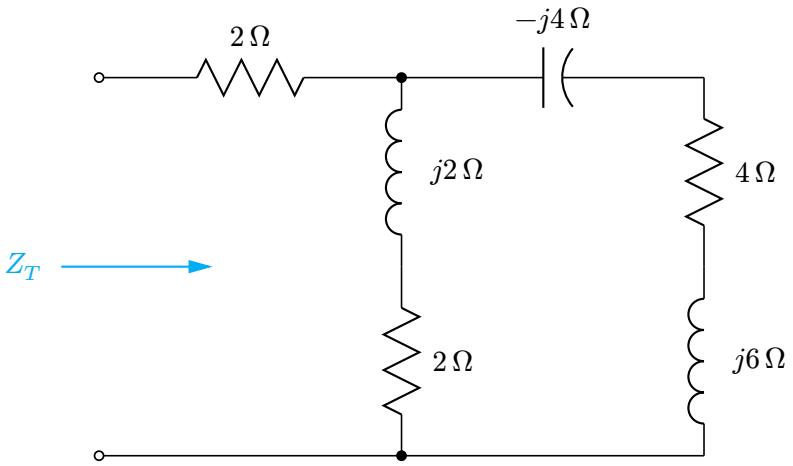
$$|\mathbf{I}| = \sqrt{\Re[\mathbf{I}]^2 + \Im[\mathbf{I}]^2} = 11.18$$
$$\theta_{\mathbf{I}} = \tan^{-1}\left(\frac{\Im[\mathbf{I}]}{\Re[\mathbf{I}]}\right) = -26.565°$$

therefore,

$$\mathbf{I} = 11.18\angle -26.565° \text{ A}$$
$$\boxed{i(t) = 11.18\cos(300t - 26.565°) \text{ A}}$$

No 2.

Step 1. Given the circuit below, compute the impedance Z_T



$$Z_{\text{right}} = -j4 + 4 + j6 = 4 + j2\,\Omega$$
$$Z_{\text{center}} = 2 + j2\,\Omega$$

$$\begin{aligned} Z_T &= (Z_{\text{right}} \parallel Z_{\text{center}}) + 2 \\ &= \left(\frac{1}{4+j2} + \frac{1}{2+j2}\right)^{-1} + 2 \\ &= \frac{18}{13} + j\frac{14}{13} + 2 \end{aligned}$$

$$\boxed{Z_T = \frac{44}{13} + j\frac{14}{13}\ \Omega \approx 3.552\angle 17.65^\circ\ \Omega}$$

Step 2. Simulate the circuit to confirm your results in Step 1.

View simulation code

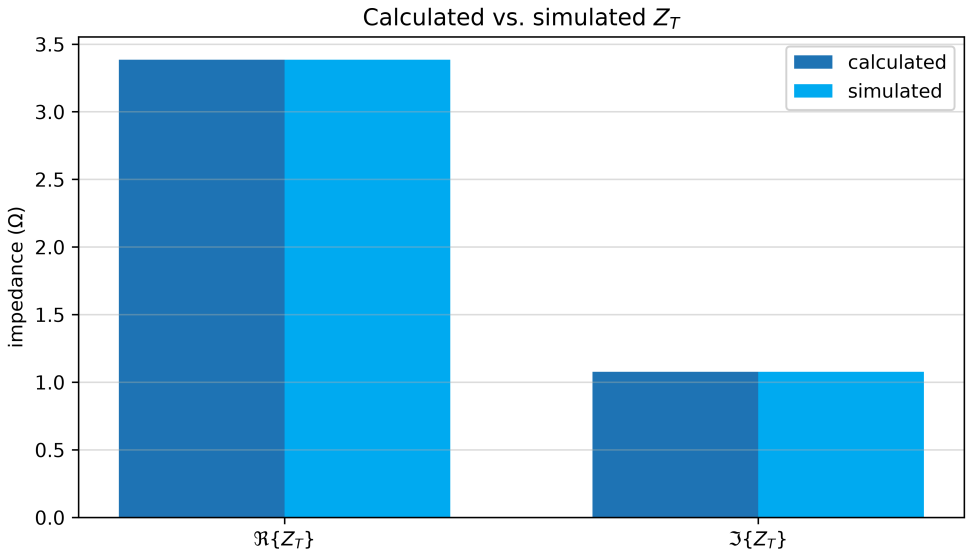
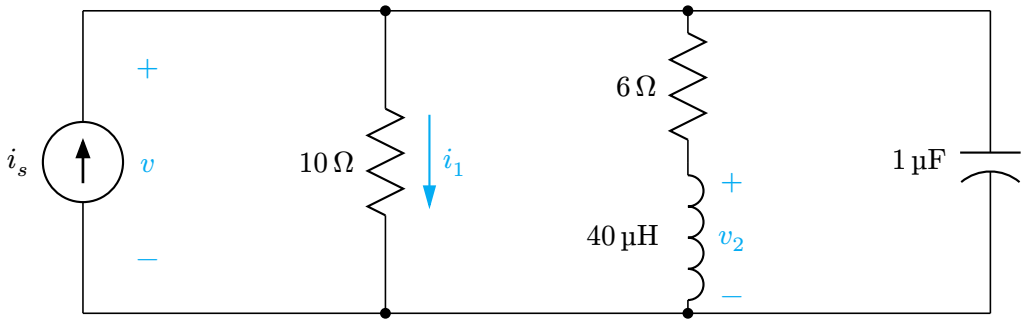


Figure 1: Comparison of the calculated and simulated real and imaginary components of the input impedance, showing agreement with $Z_T = 3.552\angle 17.65^\circ\,\Omega$.

No 3.

Step 1. Determine the time-domain expressions for steady-state value of $v(t)$, $i_1(t)$, and $v_2(t)$ for the circuit shown below.



$$i_s(t) = 8\cos(200{,}000t)\ \text{A}$$

$$\mathbf{I}_s = 8\angle 0^\circ\ \text{A}, \quad \omega = 200{,}000\ \text{rad/s}$$

$$Z_L = j8\,\Omega, \quad Z_C = -j5\,\Omega$$

$$Z_{\text{eq}} = \left(\frac{1}{Z_C} + \frac{1}{6+Z_L} + \frac{1}{10}\right)^{-1}$$

$$Z_{\text{eq}} = \left(\frac{1}{-j5} + \frac{1}{6+j8} + \frac{1}{10}\right)^{-1}$$

$$Z_{\text{eq}} = 4 - j3\,\Omega$$

$$\mathbf{V} = \mathbf{I}_s Z_{\text{eq}} \Rightarrow \mathbf{V} = 32 - j24\,\text{V} = 40\angle -36.87^\circ\,\text{V}$$

$$Z_R = \left(\frac{1}{Z_C} + \frac{1}{6+Z_L}\right)^{-1} = \frac{10}{3} - j\frac{20}{3}\,\Omega$$

$$\mathbf{I}_1 = \mathbf{I}_s\left(\frac{\frac{1}{10}}{\frac{1}{10} + \frac{1}{Z_R}}\right)$$

$$= \mathbf{I}_s\left(\frac{1}{1 + \frac{3}{1-j2}}\right)$$

$$= \mathbf{I}_s\left(\frac{1-j2}{4-j2}\right)$$

$$\mathbf{I}_1 = \frac{16}{5} - j\frac{12}{5}\,\text{A} = 4\angle -36.87^\circ\,\text{A}$$

$$\mathbf{I}_2 = \frac{\mathbf{V}}{6+j8} = -j4\,\text{A}$$

$$\mathbf{V}_2 = \mathbf{I}_2(j8) = 32\angle 0^\circ\,\text{V}$$

thus,

$$\boxed{\begin{aligned} v(t) &= 40\cos(200{,}000t - 36.87^\circ)\ \text{V} \\ i_1(t) &= 4\cos(200{,}000t - 36.87^\circ)\ \text{A} \\ v_2(t) &= 32\cos(200{,}000t)\ \text{V} \end{aligned}}$$

Step 2. What frequency will cause $i_s(t)$ and $v(t)$ to be in-phase in the steady state?

for them to be in phase, set $\Im[\Sigma Y] = 0$ and solve for ω:

$$\sum Y = \frac{1}{Z_C} + \frac{1}{6 + Z_L} + \frac{1}{10}$$

$$\sum Y = j\omega C + \frac{1}{6 + j\omega L} + \frac{1}{10}$$

$$\sum Y = j\omega C + \frac{6 - j\omega L}{6^2 + (\omega L)^2} + \frac{1}{10}$$

$$\sum Y = j\omega C + \frac{6}{6^2 + (\omega L)^2} - \frac{j\omega L}{6^2 + (\omega L)^2} + \frac{1}{10}$$

now take the imaginary parts and set their sum equal to 0:

$$0 = \omega C - \frac{\omega L}{6^2 + (\omega L)^2}$$

assuming $\omega > 0$:

$$36 + (\omega L)^2 = \frac{L}{C}$$

$$\omega = \frac{\sqrt{L/C - 36}}{L}$$

thus,

$$\boxed{\omega = 50{,}000\ \text{rad/s}}$$

Step 3. Use SPICE software to simulate the circuit and confirm your results in Steps 1-2.

View simulation code

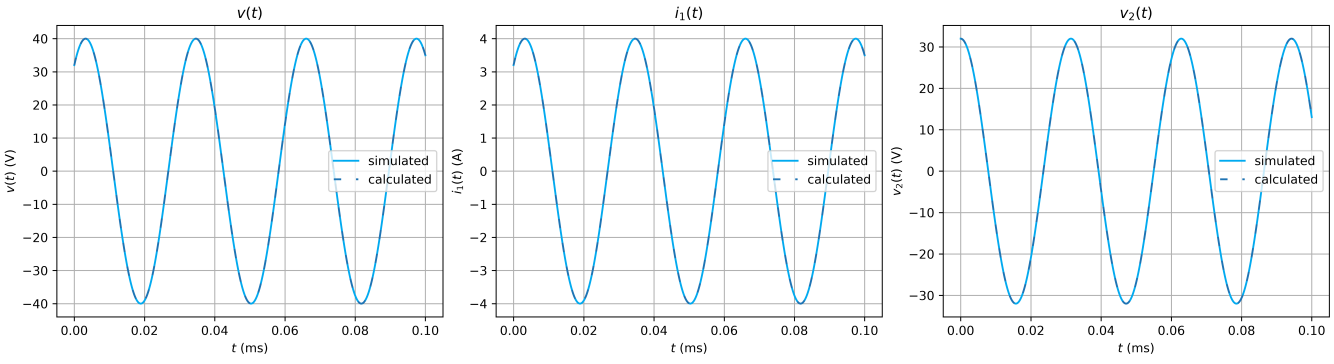
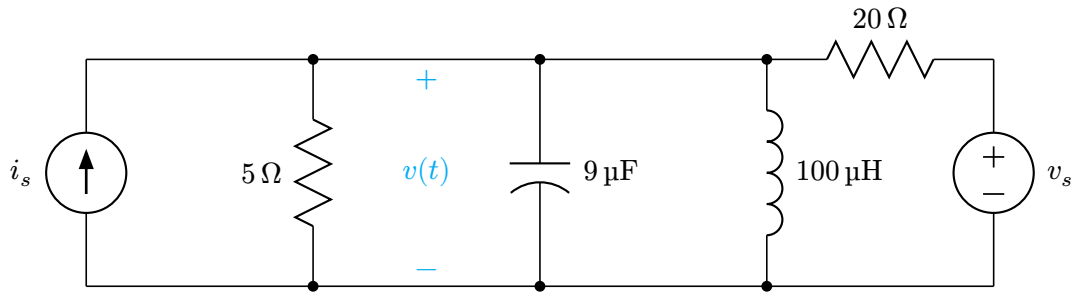


Figure 2: Comparison of the calculated and simulated steady-state waveforms for $v(t)$, $i_1(t)$, and $v_2(t)$ at $\omega = 200{,}000$ rad/s, showing agreement in amplitude and phase.

No 4.

Step 1. Determine the time-domain expression for the steady-state value of $v(t)$ for the circuit shown below.



$$i_s(t) = 10\cos(50{,}000t)\ \text{A}$$
$$v_s(t) = 100\sin(50{,}000t)\ \text{V} = 100\cos(50{,}000t - 90°)$$

$$\omega = 50{,}000t\ \text{rad/s}$$
$$\mathbf{I}_s = 10\angle 0°\ \text{A}, \quad \mathbf{V}_s = 100\angle -90°\ \text{V}$$

$$Z_L = j5\,\Omega, \quad Z_C = -j\frac{20}{9}\,\Omega$$

KCL @ top node:

$$0 = -\mathbf{I_s} + \frac{\mathbf{V}}{5} + \frac{\mathbf{V}}{Z_C} + \frac{\mathbf{V}}{Z_L} + \frac{\mathbf{V}-\mathbf{V}_s}{20}$$
$$\mathbf{I}_s = \frac{\mathbf{V}}{5} + j\frac{9\mathbf{V}}{20} - j\frac{\mathbf{V}}{5} + \frac{\mathbf{V}-\mathbf{V}_s}{20}$$
$$20\mathbf{I}_s = 4\mathbf{V} + j9\mathbf{V} - j4\mathbf{V} + \mathbf{V} - \mathbf{V}_s$$
$$20\mathbf{I}_s = \mathbf{V}(5 + j5) - \mathbf{V}_s$$

$$\mathbf{V} = \frac{20\mathbf{I}_s + \mathbf{V}_s}{5 + j5}$$

$$\begin{aligned}\mathbf{V} &= \frac{200 - j100}{5 + j5} \\ &= 20\left(\frac{2 - j}{1 + j}\right) \\ &= 10 - j30\,\text{V} \\ &= 31.623\angle -71.57°\,\text{V}\end{aligned}$$

thus,

$$\boxed{v(t) = 31.623\cos(50{,}000t - 71.57°)\,\text{V}}$$

Step 2. Use SPICE software to simulate the circuit and confirm your result in Step 1.

View simulation code

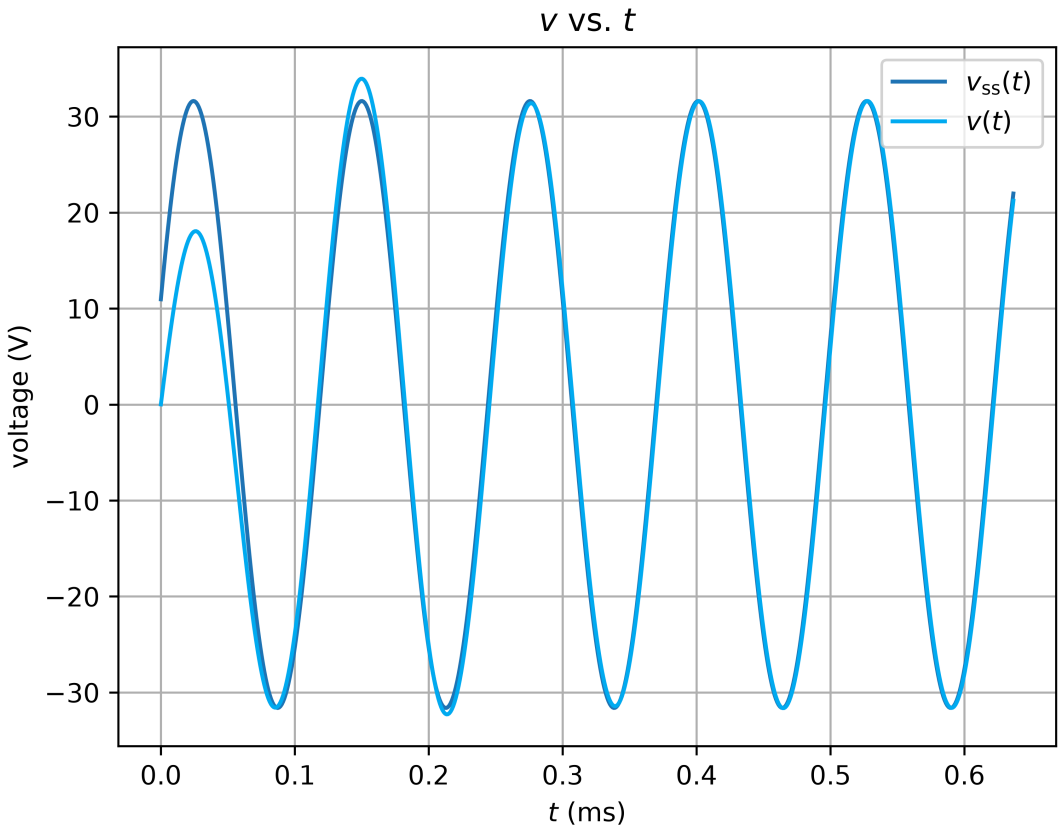


Figure 3: Comparison of the simulated voltage and analytical steady-state response, $v_{\text{ss}} = 31.623\cos(50{,}000t - 71.57°)\,\text{V}$. The initial transient decays, and the simulated waveform approaches the analytical result.